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DEPARTMENTOFCOMPUTERSCIENCE AND ENGINEERING

CCS356 - OBJECT ORIENTED SOFTWARE ENGINEERING

ASSIGNMENT - 1

TOPIC:

Railway Reservation System

Submitted By

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Railway Reservation System

1. Problem Description & System Overview:

The Railway Reservation System is a computerized information system developed to manage railway ticket booking, cancellation, passenger records, train schedules, and seat availability in an efficient and user-friendly manner.

In traditional railway reservation methods, passengers had to stand in long queues at reservation counters. This manual process often resulted in:

- Long waiting times
- Human calculation and recording errors
- Difficulty in maintaining and retrieving passenger records
- Lack of real-time seat availability information
- Inefficient cancellation and refund handling

To overcome these limitations, the Railway Reservation System introduces an **online, automated, and centralized reservation platform** accessible through web and mobile applications.

Major Services Provided by the System

- Online ticket booking and cancellation from anywhere
- Real-time seat availability and waiting list status
- Secure storage of passenger and travel data
- Automatic generation of **PNR (Passenger Name Record)**
- Digital payment through secure gateways
- Administrative control over trains, routes, and schedules

Benefits of the System

- Improves operational efficiency of railway services
- Reduces paperwork and manual errors

- Provides transparency in booking and cancellation
- Enhances passenger convenience and satisfaction
- Enables fast data retrieval and reporting

Overall, the Railway Reservation System acts as a **complete digital solution for railway ticket management**.

2. Requirements Analysis:

Requirements analysis identifies what the system must do (**functional**) and how well it must perform (**non-functional**).

A. Functional Requirements

The system should:

- Allow **new users to register** and existing users to log in securely
- Enable passengers to **search trains** using source, destination, and journey date
- Display **train schedules, seat availability, fare details, and class types**
- Allow users to **book tickets** and automatically generate a **unique PNR number**
- Support **ticket cancellation** and calculate refund according to rules
- Provide **online payment processing** via debit card, credit card, UPI, or net banking
- Maintain **passenger details, booking history, and transaction records**
- Allow **administrators** to:
 - Add new trains
 - Update routes, timings, and fares

- Delete or reschedule trains
- View booking statistics and reports

B. Non-Functional Requirements

The system must also ensure:

- **High performance** with fast response time during peak booking hours
- **Security** through authentication, authorization, and encrypted payments
- **Data accuracy and integrity** in all transactions
- **User-friendly interface** for easy navigation by passengers of all age groups
- **24×7 system availability** with minimal downtime
- **Backup and disaster recovery** to prevent data loss
- **Scalability** to handle millions of users simultaneously

These requirements guarantee that the system is **reliable, secure, and efficient**.

3. Use Case Diagram :

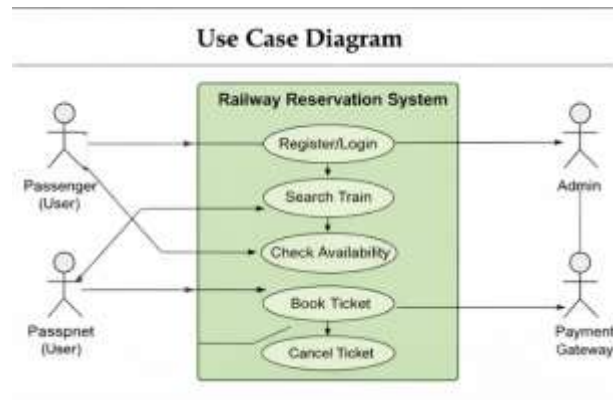
Actors Involved

1. **Passenger/User** – books and cancels tickets
2. **Admin** – manages trains, schedules, and system data
3. **Payment Gateway** – processes financial transactions

Primary Use Cases

- User Registration and Login
- Train Search
- Seat Availability Check

- Ticket Booking
- Online Payment
- Ticket Cancellation and Refund
- Train Management (Admin side)



Use Case Interaction Description

The passenger logs into the system, searches for trains, checks availability, and proceeds to booking.

During booking, the system interacts with the **payment gateway** to complete the transaction securely.

After successful payment, the system confirms the reservation and generates a **PNR**.

The **admin** maintains backend operations such as updating schedules and monitoring bookings.

This diagram represents the **overall functional behavior and user interaction flow** of the system.

4. Class Diagram :

The class diagram shows the **static structure** of the system.

Main Classes and Responsibilities

1. User Class

- Attributes: userID, name, email, password, phone
- Functions: register(), login(), logout()

2. Train Class

- Attributes: trainID, trainName, source, destination, schedule, fare
- Functions: getSchedule(), updateSchedule()

3. Ticket Class

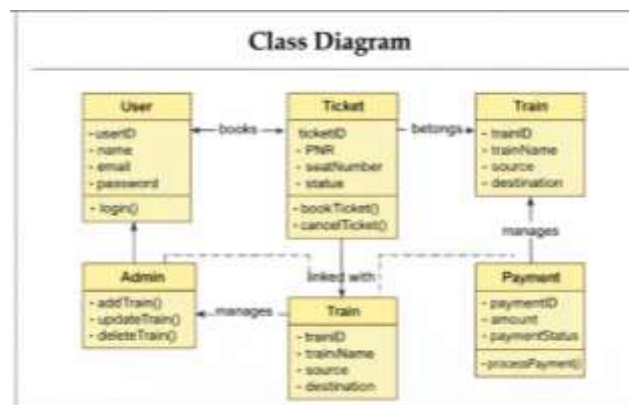
- Attributes: ticketID, PNR, seatNumber, travelDate, status
- Functions: bookTicket(), cancelTicket(), generatePNR()

4. Payment Class

- Attributes: paymentID, amount, paymentStatus, paymentMode
- Functions: processPayment(), refundPayment()

5. Admin Class

- Functions:
 - addTrain()
 - updateTrain()
 - deleteTrain()
 - viewReports()



Relationships Between Classes

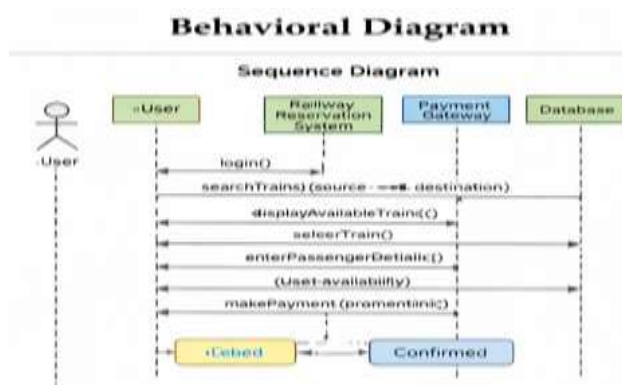
- A User books Ticket
- A Ticket is associated with one Train
- Payment confirms Ticket booking
- Admin manages Train data

This structure ensures **clear data organization and maintainability**.

5. Behavioral Diagram :

Behavioral diagrams describe **how the system behaves over time**.

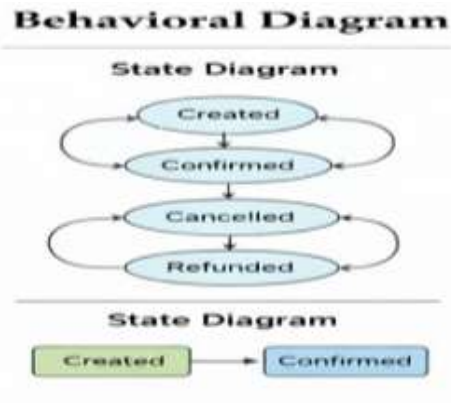
A. Sequence of Ticket Booking



1. User logs into the system
2. Searches trains using journey details
3. System retrieves and displays available trains
4. User selects train and enters passenger information
5. System checks seat availability in database
6. User proceeds to secure payment
7. Payment gateway verifies transaction
8. System confirms booking and generates **PNR**

9. Ticket details are stored and shown to the user

B. Ticket State Transitions



- **Created** – ticket generated but not confirmed
- **Confirmed** – payment successful and seat reserved
- **Cancelled** – user cancels the ticket
- **Refunded** – refund processed to user account

This represents the **complete lifecycle of a railway ticket**.

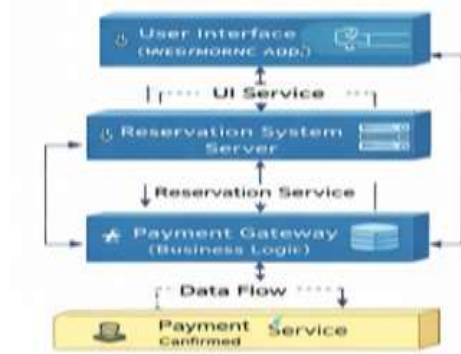
6. Component and Deployment Diagram:

A. Component Diagram:

Major software components:

- **User Interface** – web/mobile screens for users and admins
- **Reservation/Application Server** – handles business logic and booking process
- **Database Server** – stores trains, users, tickets, and payments
- **Payment Gateway API** – manages secure financial transactions

Component Diagram



Workflow:

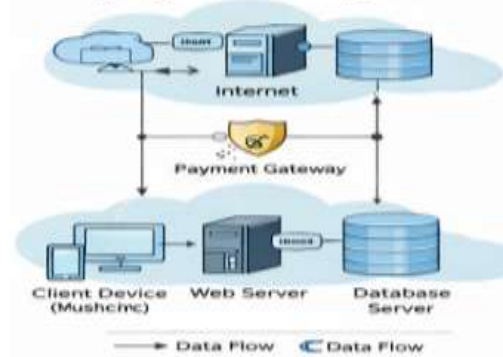
UI → Application Server → Database → Payment Gateway → Confirmation to User.

B. Deployment Diagram:

Physical system nodes:

- Client Device (mobile or computer)
- Web Server
- Application Server
- Database Server
- External Payment Gateway

Deployment Diagram



Data Flow:

Client → Internet → Web Server → Application Server → Database →
Response back to Client.

Advantages of This Deployment

- Supports **large-scale users**
- Ensures **data security and centralized control**
- Provides **high availability and reliability**

Conclusion:

The Railway Reservation System is a **modern, automated, and scalable solution** that simplifies railway ticket booking and management.

By integrating **real-time databases, secure payments, and user-friendly interfaces**, the system enhances both **passenger experience and railway administration efficiency**.